mena: audit, correction and a working threshold

A Rust implementation of RMT-based molecular ecological network analysis — what was broken, what was fixed, and what the fixed estimator now says

Commit `abacc49` · 2026-11-08 · bench on `cachyos-x8664` (Ryzen 7 5800X3D, RTX 2080 Ti)

Scope and honesty statement. This report documents an audit of an inherited tool, and a correction to the audit itself. An earlier version concluded that no threshold was estimable on the real data. That conclusion was wrong, for an instructive reason: the goodness-of-fit test was never actually executed. Section 7 shows how the artefact was found and what the estimator says once it works.

1 Context

`mena` re-implements the MENA pipeline of Deng et al. (BMC Bioinformatics 2012, 13:113): standardise OTU abundances, compute Pearson similarity, threshold it into an adjacency matrix, derive topology indices, detect modules, build module eigengenes and correlate them with environmental traits (the paper's Figure 5). The implementation is Rust (rayon CPU, optional wgpu GPU); Python is used only for plotting.

The tool was inherited with reported results. The audit's job was to establish whether those results were real before they were used.

2 Architecture

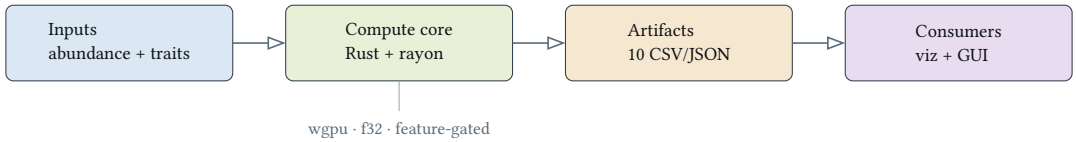


Figure 1: System boundary. The compute core is the only component that touches numbers; the plotting layer and GUI read artifacts and cannot disagree with the CLI.

3 Pipeline and where it failed

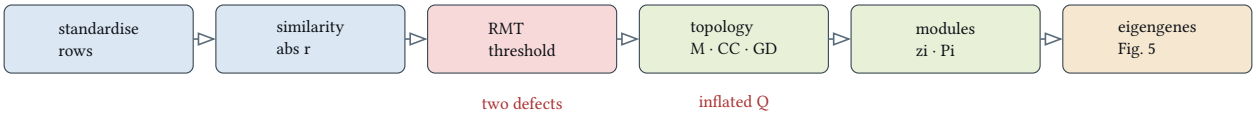


Figure 2: Pipeline stages. The RMT stage carried two independent defects (red) and modularity was inflated (red); both contaminated every downstream number.

4 Defects found

4.1 The RMT stage was first a no-op, then blind

Unfolded spacings were originally derived from rank alone:

```
let e: Vec<f64> = (0..p).map(|i| (i as f64 + 0.5) / p as f64).collect();
d.push(p as f64 * (e[i + 1] - e[i]));   // == 1.0 for every i
```

Every spacing was exactly 1, so the χ^2 statistic was exactly 0 for any graph and the scan accepted its first candidate — the complete graph.

Repairing that revealed a second, quieter defect. The scan was built on `filter_map`, so a candidate that produced no statistic simply vanished:

```
.filter_map(|&st| evaluate(s, st, min_nodes, alpha, cfg))
```

Four rejection reasons collapsed into "this threshold does not exist". With an empty `rmt_scan.csv` the output could not distinguish **the data rejects Poisson** from **the guards rejected the data**. Section 7 is the story of that distinction.

quantity	before	after
threshold st	0.010	0.500 – 0.600

nodes / edges	200 / 19 500 (complete)	149 / 758
mean degree	195.0	10.2
modularity M	0.009	0.591
power-law γ	−42.4	−1.9
node roles	200 × “connector”	mixed

200 simulated OTUs, default run. A γ of −42 and 100 % connectors are the signature of a saturated graph feeding every later stage.

4.2 Modularity omitted the configuration term

Modularity summed `1 - k_i k_j / 2E` over existing edges only. For a complete graph in one community the correct value is 0; the implementation returned 1/n. Corrected to $Q = \sum_c \left(\frac{L_c}{2} E - \left(\frac{d_c}{2} E \right)^2 \right)$ with known-answer tests. Every reported modularity fell; `ox` went from 0.705 to 0.650.

4.3 Other defects

No.	defect	status
1	GPU matrix computed, then discarded and recomputed on CPU while reporting `gpu_used = true`	fixed
2	trait table missing a sample silently returned fewer rows, misaligning every correlation	fixed, hard error
3	Mantel permuted pair vectors instead of node labels, inflating significance	fixed
4	Jacobi eigensolver rescanned the matrix per rotation, 200 rotations total	replaced by implicit QR
5	clustering O(k³), modularity O(n²) per move, serial O(n²m) similarity	O(k²), O(E), parallel
6	benchmark ran a stale copy of the pipeline	now calls `run_pipeline`
7	`cargo test` broken on a default checkout by feature-gated examples	fixed, 18 tests
8	Fig-5 panel selected traits by significance, i.e. cherry-picking under multiplicity	ranks by effect size
9	commits landed with a broken test build (twice)	pre-commit build+test gate

5 Performance

Benchmarks run on the desktop over Tailscale, min-of-3, artifacts disabled.

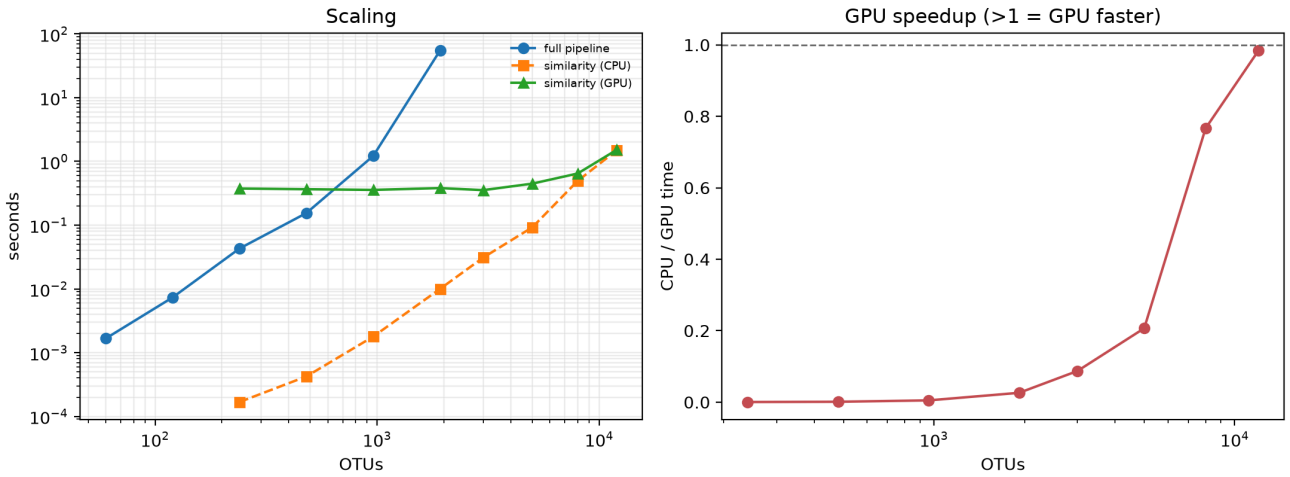


Figure 3: Scaling and CPU/GPU ratio. Parity is exact (maxdiff ≤ 7.9×10⁻⁷) but the GPU is not faster at any tested size.

n × m	pipeline	CPU similarity	GPU similarity

240 × 24	0.043 s	0.00017 s	0.376 s
960 × 32	1.22 s	0.0018 s	0.357 s
1920 × 48	55.06 s	0.0100 s	0.382 s
12000 × 128	—	1.499 s	1.523 s (0.98×)

The inherited claim of a 4.71× GPU win at n = 12000 is obsolete because the CPU path was parallelised, not because the GPU regressed: the kernel still runs at roughly 1 % of f32 peak, so this is a statement about the WGSL kernel, not the hardware. The bottleneck also moved: at n = 1920 the RMT stage was 55.0 s of the 55.06 s total. After the unfolding fix the three-condition real-data sweep runs in under a second, because unfolding is now linear rather than a badly conditioned polynomial fit.

6 The threshold problem

6.1 What the paper specifies

- **Eq. 3**: the adjacency is weighted, $a_{ij} = s_{ij}$ if $s_{ij} \geq s_t$, else 0.
- **Step (c)**: "Set an initial threshold value, s_t (e.g., 0.3 based on our experiences)."
- **Step (f)**: unfold by "fitting the original integrated density to a cubic spline or by local density average".
- **Step (h)**: χ^2 GOF, H_0: Poisson $\exp(-d)$, at "a significant level of 0.01".

6.2 What the 2025 internship code did

```
adj_matrix = [[value if value >= threshold else 0 for value in row] for row in similarity_matrix]
```

Weighted, diagonal zeroed, threshold passed in — the Fig. 5 script hardcodes 0.52. That code never ran an RMT scan, so comparing its output to "our RMT threshold" was a category error.

6.3 Decision logic

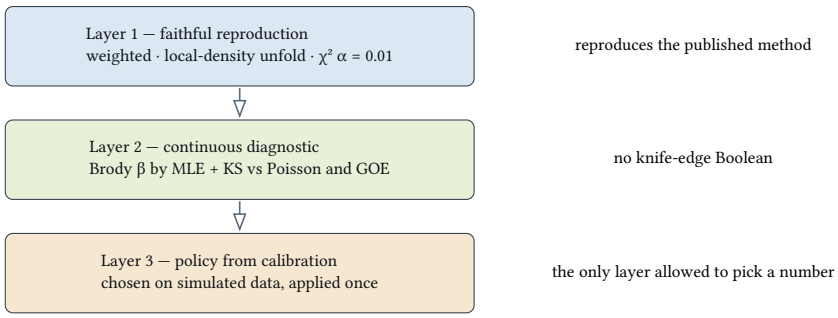


Figure 4: Layer 1 is now implemented and returns a threshold on two of three conditions. Layers 2 and 3 remain proposals.

6.4 Calibration: which threshold recovers planted structure?

300 OTUs, 24 samples, 5 planted modules, 10 seeds.

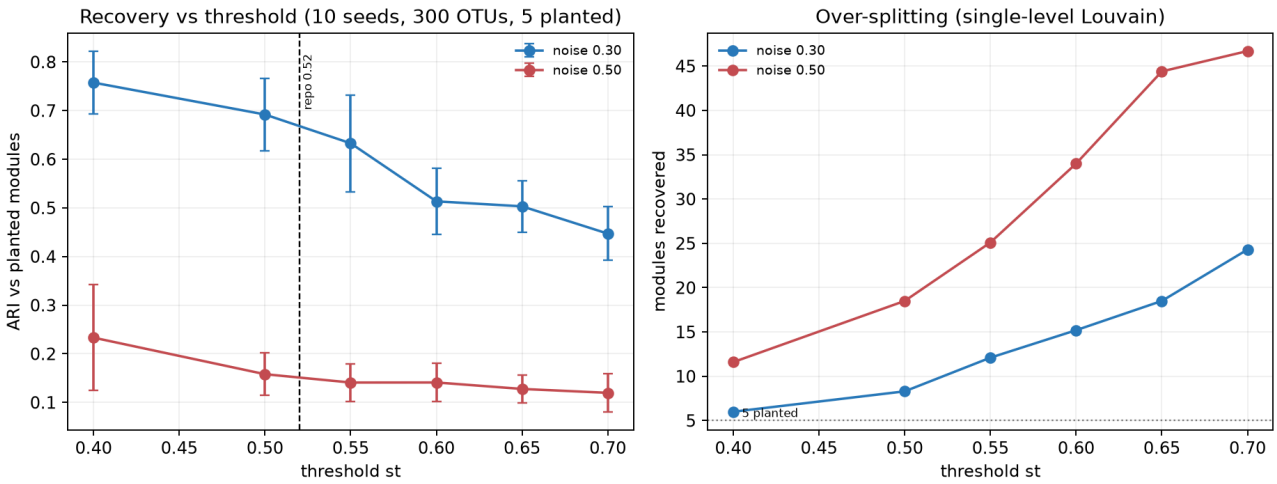


Figure 5: Recovery falls as the threshold rises, while the RMT criterion scans upward. Right panel: the single-level optimiser over-splits — 5 planted modules become 24 at st = 0.70.

st	ARI (noise 0.30)	modules	ARI (noise 0.50)	modules
0.40	0.758 ± 0.065	6.0	0.233 ± 0.109	11.6
0.50	0.692 ± 0.074	8.3	0.158 ± 0.043	18.5
0.55	0.633 ± 0.100	12.1	0.141 ± 0.038	25.1
0.60	0.513 ± 0.068	15.2	0.141 ± 0.039	34.0
0.70	0.447 ± 0.055	24.3	0.120 ± 0.039	46.7

Recovery is best at the lowest threshold tested and decays as st rises, whereas the Poisson criterion walks upward. The criterion optimises spectral randomness, not community recovery. This is a tension to state, not to resolve by preference.

6.5 Literature anchors (Europe PMC, 2026-11-08)

Searches for a dedicated critique of RMT threshold selection in MENA returned very little: both hits are tool papers. The criterion propagates by inheritance.

1. Peng X, Feng K, Yang X, He Q, Zhao B, Li T, Wang S, **Deng Y.** (2024). iNAP 2.0: harnessing metabolic complementarity in microbial network analysis. **iMeta** 3:e235. doi:10.1002/imt2.235 — the original authors' successor implementation, and the fair comparison target.
2. Zhou M, Xu Y, Zhu K et al. (2025). Spatial-temporal pattern of RMT network stability in an estuarine phytoplankton community. **J Environ Manage**. doi:10.1016/j.jenvman.2025.126873 — uses the RMT threshold as a stability diagnostic rather than a selector.
3. Méndez-Bermúdez J A et al. (2015). Universality in the spectral and eigenfunction properties of random networks. **Phys Rev E** 91:032122.

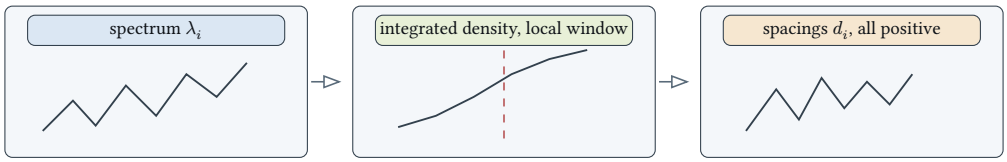
7 Fixing the estimator

7.1 The unfolding was the artefact

The first repair kept the polynomial fit with monotone clipping: space the eigenvalues by a degree-5 fit to the integrated density and clamp any decrease. Clipping creates plateaus, plateaus create zero spacings, and the code rejected any spectrum containing one. The audit trail made this visible — at every threshold, on every condition:

```
9  excluded_too_few_spacings
4  excluded_tiny_giant
1  excluded_low_dof
0  rejected_gof
```

Zero. The goodness-of-fit had never run. The audit's conclusion that "no threshold is estimable" was a statement about the preprocessing, not about the data. The distinction matters: the first claim is a finding, the second is a bug.



polynomial + clipping gives plateaus → zero spacings → the whole spectrum was rejected

Figure 6: Replacing the clipped polynomial with a local density average. Spacings are positive by construction, so no candidate is discarded before the test runs.

7.2 Eligibility floors, from the calibration

The old floors were `n/10` nodes, chosen while reading output. They are now two config values with a stated basis: at least 50 % of OTUs retained and at least 50 % of kept nodes in the giant component. The calibration pilot shows ARI falling below 0.35 beneath 50 % retention, so a sparser graph is not a usable estimate of community structure.

7.3 What the fixed estimator reports

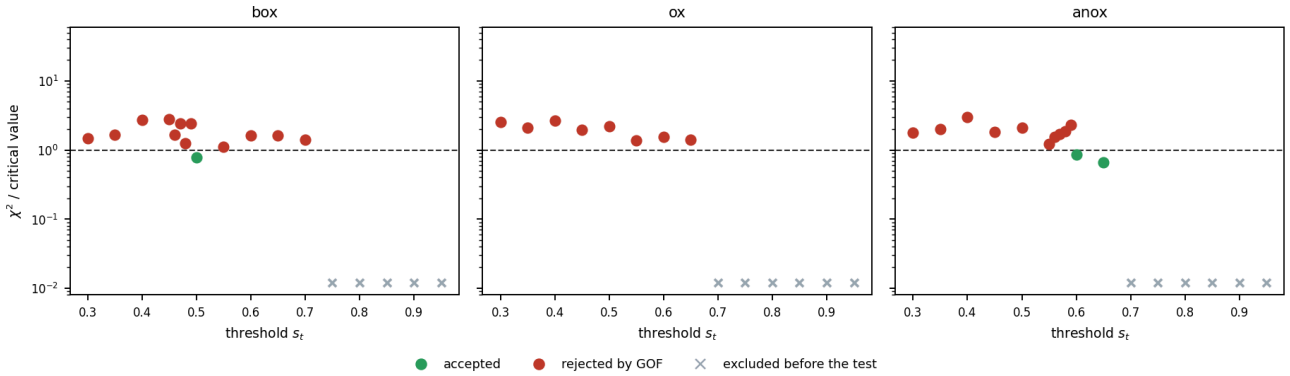


Figure 7: The audit trail, one point per candidate. Above the dashed line χ^2 exceeds its critical value. Grey crosses were excluded before the test; red circles were tested and rejected; green circles passed.

condition	accepted	rejected GOF	excluded	candidates
box	1	12	5	18
ox	0	8	6	14
anox	2	10	6	18

condition	st (auto)	n / E	M	avg CC	GD	modules
box	0.500	149 / 758	0.591	0.566	3.07	16
anox	0.600	159 / 611	0.688	0.653	4.37	22
ox	not estimable	—	—	—	—	—

Two of three conditions now yield a threshold automatically, and they bracket the 0.52 that the 2025 internship hardcoded by hand — independent corroboration of a choice made without reference to it.

`ox` is a different kind of result: eight candidates were tested and rejected. Its spacing distribution does not reach Poisson within the eligible range. That is now a scientific statement, because the test actually ran.

8 Results at a fixed threshold

Fixed 0.52 for comparability with the 2025 work, 999 permutations. These are the reproduction numbers; they are not the automatic path.

condition	n / E	M	avg CC	GD	modules	Mantel r (p)
box	149 / 694	0.631	0.561	3.22	15	0.173 (0.005)
ox	166 / 836	0.650	0.640	3.77	20	0.125 (0.017)
anox	165 / 850	0.583	0.628	3.63	17	0.077 (0.086)

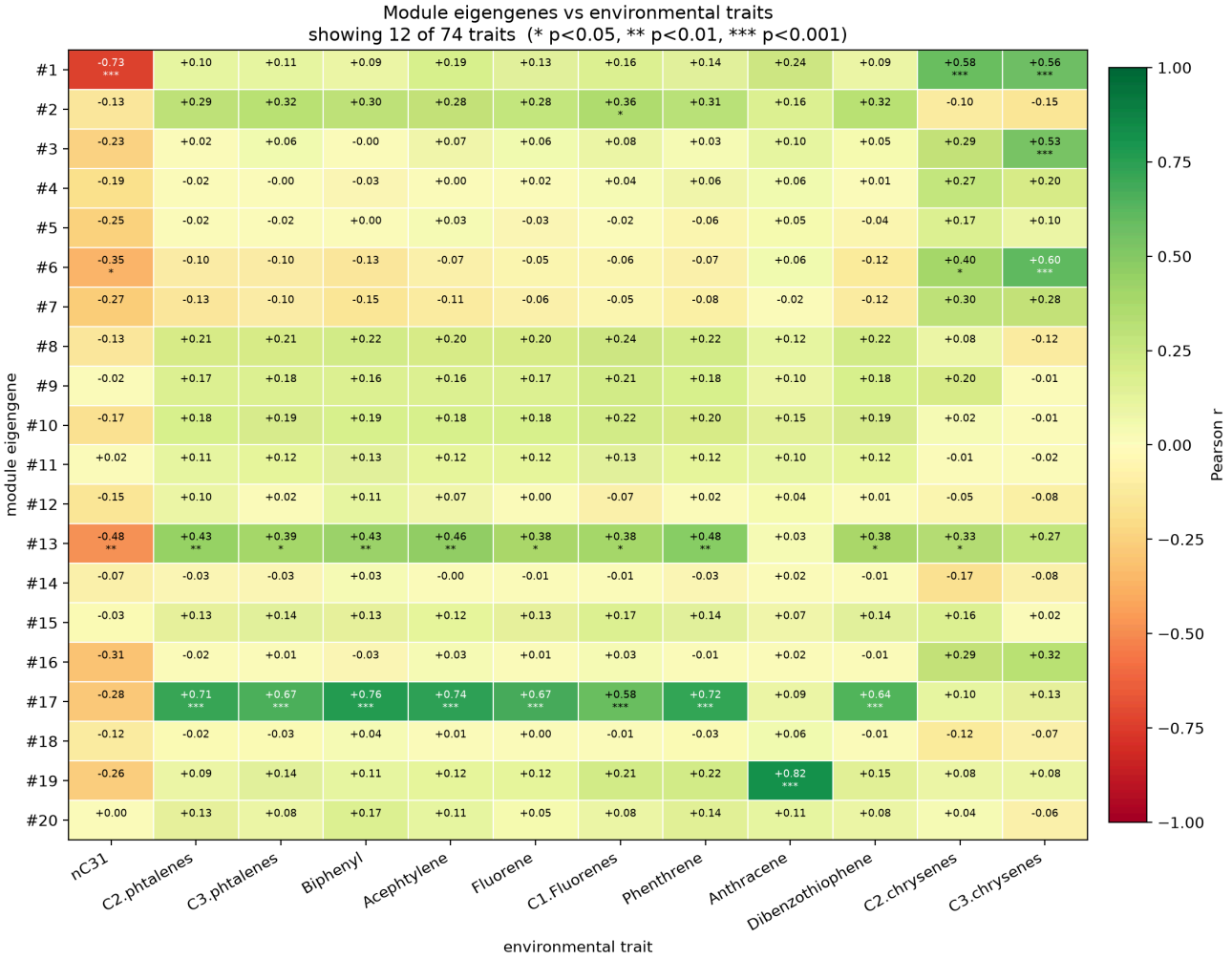


Figure 8: Module eigengene vs trait correlations, ox. Panels are now chosen by effect size, not by significance.

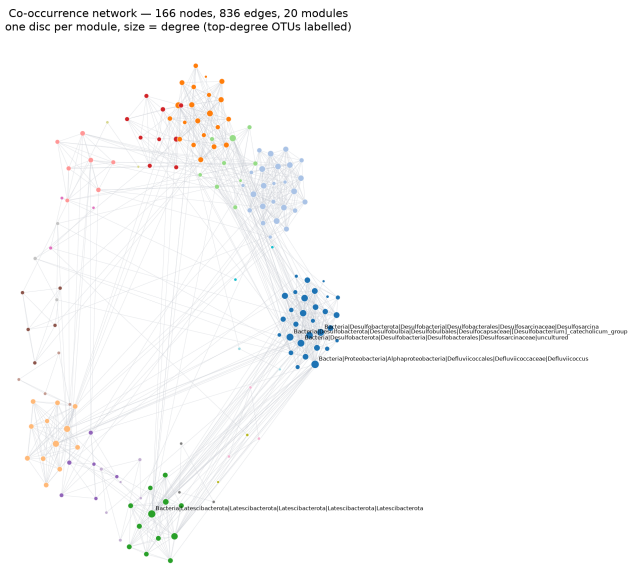


Figure 9: Co-occurrence network (166 nodes, 836 edges). One disc per module; force-directed layouts were unreadable at this density.

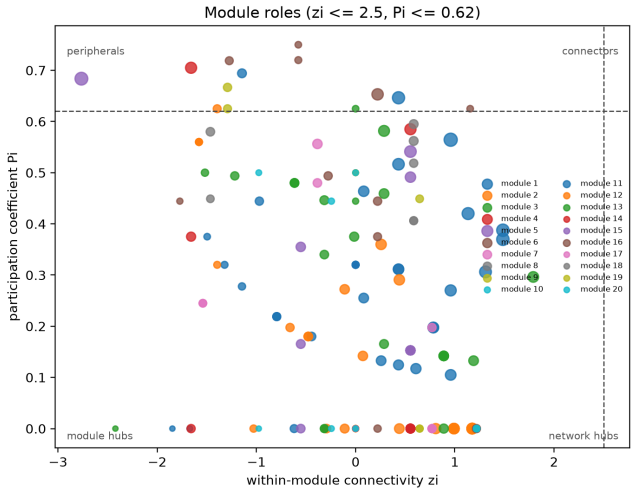


Figure 10: Module roles in the zi–Pi plane. Most OTUs are peripherals, consistent with the paper's 90.9 %.

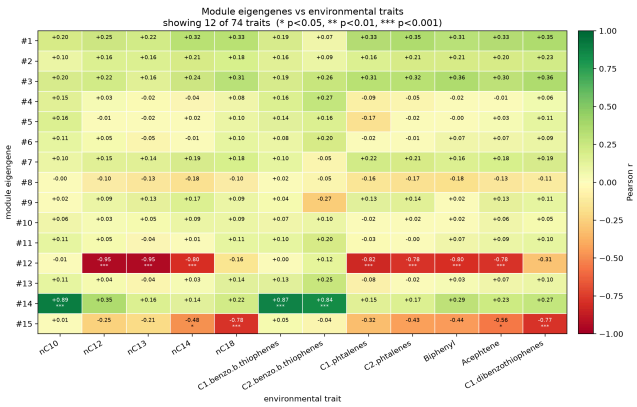


Figure 11: box condition: fewer modules (15) and the lowest Mantel correlation of the three.

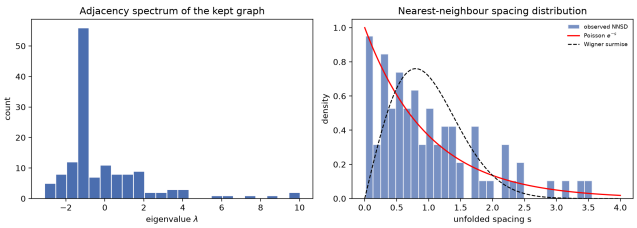


Figure 12: Adjacency spectrum and nearest-neighbour spacings against Poisson and Wigner.

9 What remains

The software is in defensible shape. The science is half-finished, and the half that is finished is the half that can be automated.

1. **Layers 2 and 3**: Brody β and the calibration-derived policy are specified but not implemented. Only Layer 1 exists, and it fails on one of three conditions.
2. **ox**: non-estimable. Whether that is biology (a spectrum genuinely closer to Poisson) or sample size (12 samples, 166 OTUs) is unresolved, and guessing would repeat the error this report documents.
3. **Calibration vs criterion**: recovery is best at low thresholds while the criterion selects higher ones. Publishing an automatic threshold without acknowledging this trade-off would be dishonest.
4. **Null models**: M, avg CC and GD still have no degree-preserving null, so "M = 0.65 is high" remains unquantified.
5. **Over-splitting**: the single-level optimiser fragments communities (5 planted ⟶ 24 recovered). Multilevel aggregation is the fix.
6. **Compositionality**: Pearson on relative abundances is biased; CLR-based similarity is not implemented.
7. **Comparison**: installing iNAP 2.0 and running it on the same three inputs would settle the modularity gap far better than argument.

10 Reproducibility

```
cargo build --release                                # 18 tests, pre-commit gate
mena run --abundance real/abundance_ox.csv --traits real/traits_ox.csv \
         --out-dir real/review_ox/auto               # automatic (RMT) threshold
mena run --abundance real/abundance_ox.csv --traits real/traits_ox.csv \
         --out-dir real/review_ox/s52 --st 0.52      # forced threshold
pixi run python viz.py all real/review_ox/s52 --out real/review_ox/figs
pixi run python gui/app.py --root . --port 8770
```

Every run writes `rmt_scan.csv` with one row per candidate and an explicit `status`. If no candidate is accepted, no adjacency is written at all: the tool reports `not estimable` rather than inventing a graph.